FP3_Presentation

February 10, 2026

FP3 – Integration Benchmarks

Comparison of numerical integration methods: Gauss–Legendre, Clenshaw–Curtis and trapezoidal rules on a set of benchmark functions.

Task 1 – Theory

Trapezoidal rule

The trapezoidal rule approximates the definite integral

$$\int_a^b f(x)\ dx$$

by partitioning the interval $[a, b]$ into N subintervals of equal length $h = \frac{b-a}{N}$. On each subinterval, the function f is approximated by a linear function, leading to the quadrature formula

$$\int_a^b f(x)\ dx \approx \sum_{k=1}^{N} \frac{f(x_{k-1}) + f(x_k)}{2} h, \quad \text{where } x_k = a + kh.$$

Two-dimensional trapezoidal rule

In two dimensions, the trapezoidal rule is obtained as a tensor product of the one-dimensional formula. For a rectangular domain $[a, b] \times [c, d]$, it takes the form

$$\int_a^b \int_c^d f(x, y), dy, dx \approx \frac{hk}{4} \Bigg(f(a, c) + f(b, c) + f(a, d) + f(b, d) + 2 \sum_{i=1}^{n-1} [f(x_i, c) + f(x_i, d)] + 2 \sum_{j=1}^{n-1} [f(a, y_j) + f(b, y_j)] + 4 \sum_{i=1}^{n-1} \sum_{j=1}^{n-1} f(x_i, y_j) \Bigg),$$

where

$$h = \frac{b-a}{n}, \qquad k = \frac{d-c}{n}, \quad x_i = a + ih, \quad y_j = c + jk.$$

Source: Simpson's Rule in Two Dimensions, IEEE CSE KTU

Gauss–Legendre quadrature

Gauss–Legendre quadrature is a Gaussian quadrature method for approximating definite integrals by evaluating the function at optimally chosen nodes (the zeros of the Legendre polynomials) with corresponding weights. On the reference interval $[-1, 1]$, the method is defined as

$$\int_{-1}^{1} f(x)\ dx \approx \sum_{i=1}^{n} w_i f(x_i),$$

where n denotes the number of quadrature points, x_i are the roots of the Legendre polynomial $P_n(x)$ and w_i are the associated quadrature weights.

Gauss–Legendre quadrature is exact for all polynomials of degree up to $2n - 1$ and provides very high accuracy for smooth functions.

Two-dimensional Gauss–Legendre quadrature

In higher dimensions, Gauss–Legendre quadrature is constructed via tensor products of the one-dimensional rule. In two dimensions, the resulting formula on the square $[-1, 1]^2$ is

$$\int_{-1}^{1}\int_{-1}^{1} f(x, y)\ dy\ dx \approx \sum_{i=1}^{n}\sum_{k=1}^{n} w_i w_k f(x_i, y_k).$$

Source: Guermond, *Numerical Integration*, Texas A&M University

Clenshaw–Curtis quadrature

Clenshaw–Curtis quadrature approximates the integral of f on $[-1, 1]$ by intergrating its interpolating polynomial at Chebyshev points. These nodes are defined as

$$x_k = \cos\left(\frac{k\pi}{n}\right), \quad k = 0, \dots, n.$$

The interpolant p_n of f in these nodes satisfies

$$\int_{-1}^{1} f(x)\ dx \approx \int_{-1}^{1} p_n(x)\ dx = \sum_{k=0}^{n} f(x_k) w_k,$$

where the quadrature weights are defined by

$$w_k := \int_{-1}^{1} l_k(x) dx, \quad l_k(x) = \prod_{\substack{j=0 \\ j\neq k}}^{n} \frac{x - x_j}{x_k - x_j}.$$

However this formula can also have easier to implement form if we take into consideration that

$$p_n(x) = \sum_{k=0}^{n} a_k T_k(x), \quad T_k(x) = cos(k \arccos(x)).$$

The Chebyshev polynomials satisfy

$$\int_{-1}^{1} T_k(x)\ dx = \begin{cases} \dfrac{2}{1-k^2}, & k \text{ even}, \\ 0, & k \text{ odd}. \end{cases},$$

so that

$$\int_{-1}^{1} p_n(x)\ dx = \sum_{k=0}^{n} a_k \int_{-1}^{2} T_k(x)\ dx$$

reduces to a simple linear combination of the coefficients a_k.

These coefficients are easily computed from the samples $f(x_j)$ via the discrete cosine transform

$$a_k = \frac{f(1)}{n} + \frac{(-1)^k f(-1)}{n} + \frac{2}{n} \sum_{j=1}^{n-1} f\left(\cos\left(\frac{j\pi}{n}\right)\right) \cos\left(\frac{jk\pi}{n}\right)$$

Source: Clenshaw, C.W., Curtis, A.R. A method for numerical integration on an automatic computer. Numer. Math. 2, 197–205 (1960)

Clenshaw–Curtis quadrature exhibits rapid convergence for smooth functions and, despite relying on non-optimal nodes in the Gaussian sense, is often competitive with Gauss–Legendre quadrature in practical applications.

Two-dimensional Gauss–Legendre quadrature

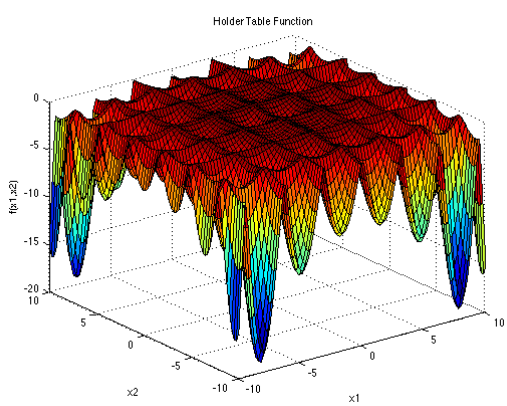
In higher dimensions, Clenshaw–Curtis quadrature is - as in Legendre-Gauss case - constructed via tensor products of the one-dimensional rule.

Task 2 – Benchmark Integrals

Adnotation: To avoid focusing on the specific properties of individual functions and instead emphasize the quality of their numerical approximation, we restrict our analysis to two-dimensional functions defined on the domain $[-1, 1]^2$. The functions that will be used for this examination are:

Holder Table Function

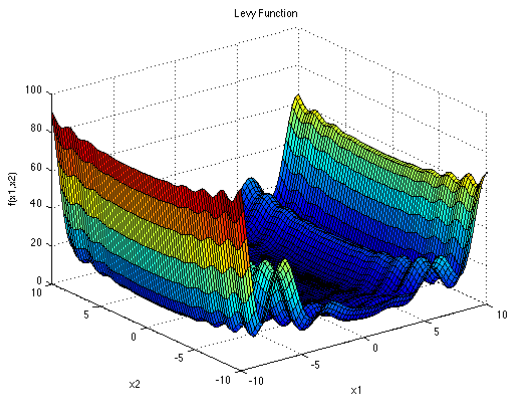
Type of function: oscillatory



$$f(x_1, x_2) = -\left|\sin(x_1)\cos(x_2)\exp\left(\left|1 - \frac{\sqrt{x_1^2 + x_2^2}}{\pi}\right|\right)\right|$$

Levy Function

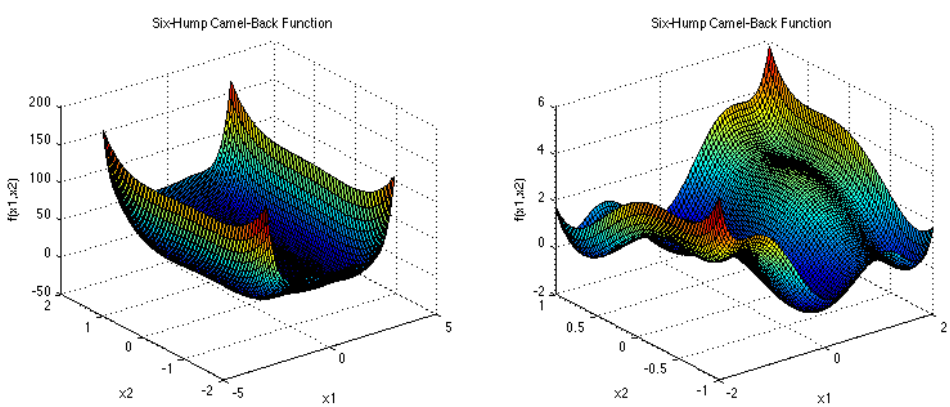
Type of function: smooth, oscillatory.



$$f(x_1, x_2) = \sin^2(\pi w_1) + (w_1 - 1)^2 \left[1 + 10\sin^2(\pi w_1 + 1)\right] + (w_2 - 1)^2 \left[1 + \sin^2(2\pi w_2)\right], \qquad w_i = 1 + \frac{x_i - 1}{4}$$

Six-Hump Camel Function

Type of function: smooth, rational.



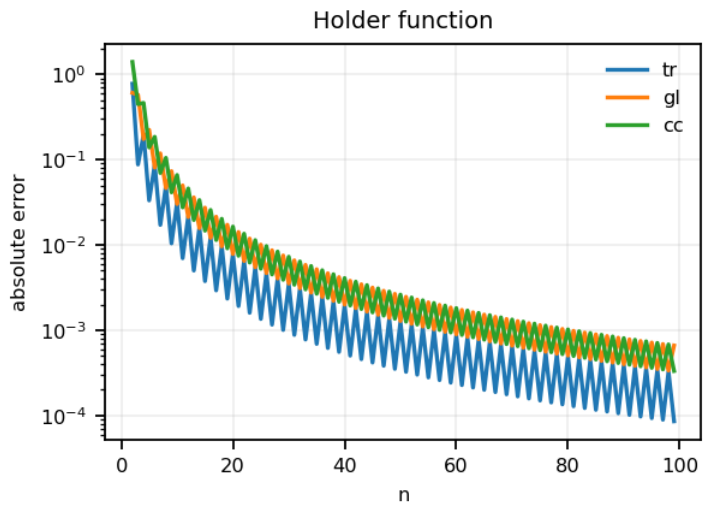
$$f(x_1, x_2) = \left(4 - 2.1x_1^2 + \frac{x_1^4}{3}\right) x_1^2 + x_1 x_2 + \left(-4 + 4x_2^2\right) x_2^2$$

Task 3 – Implementation

Code with implementation of Trapezoidal rule, Gauss–Legendre quadrature and Clenshaw–Curtis quadrature is included in the file .ipynb

Task 4 – Comparison

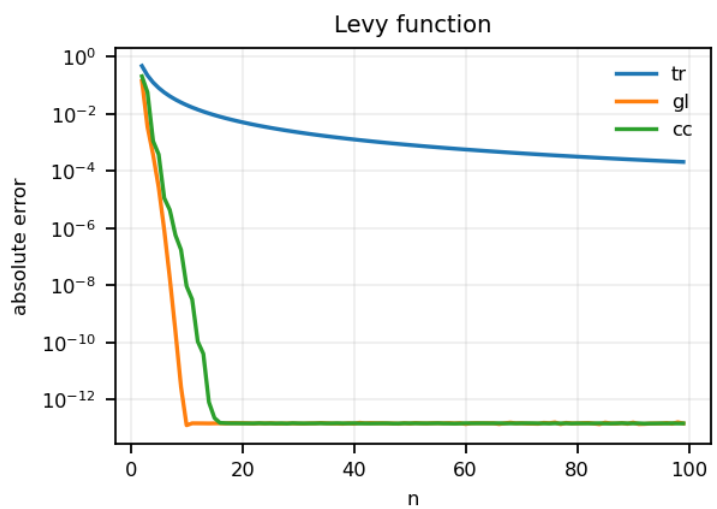
Compare accuracy and convergence as the number of nodes increases.



Reminder Holder function is oscillatory.

The trapezoidal rule achieves the smallest errors for large values of n despite its slower theoretical convergence rate as we can see for the highest n. Gauss–Legendre and Clenshaw–Curtis initially converge faster, but their accuracy stagnates and remains above that of the trapezoidal rule as n increases. This behavior is caused by the strong oscillations and lack of smoothness of the Holder function, which reduce the effectiveness of numerical methods.

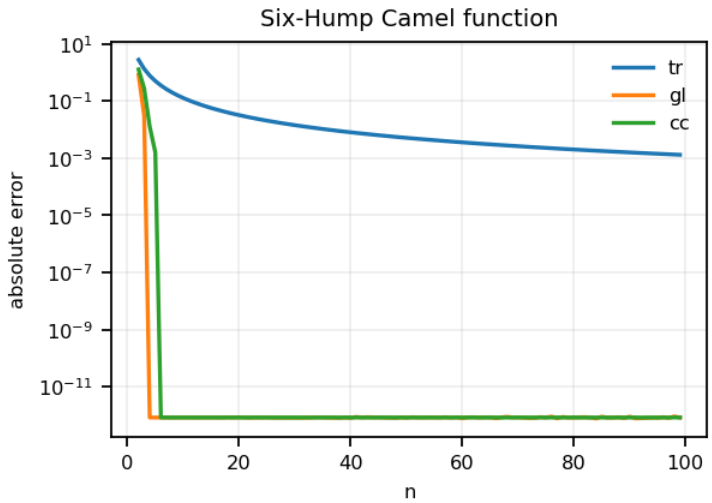
As a result, the trapezoidal rule outperforms both Gauss–Legendre and Clenshaw–Curtis quadrature for sufficiently large numbers of nodes in this specific case.



Reminder Levy function is smooth and oscillatory.

Gauss–Legendre and Clenshaw–Curtis errors drop rapidly already for small $n \approx 15$, reaching a near machine-precision plateau around 10^{-13}. In contrary, the trapezoidal rule converges much slower and remains a lot less accurate across the entire range of n. Gauss–Legendre attains the plateau slightly earlier, while Clenshaw–Curtis follows closely and matches the same accuracy level.

To conclude, both Gauss–Legendre and Clenshaw–Curtis vastly outperform the trapezoidal rule in both accuracy and convergence speed.



Reminder Six-Hump Camel function is smooth and rational.

As Six-Hump Camel function is also smooth, Gauss–Legendre and Clenshaw–Curtis quadrature again converge extremely rapidly, reaching machine-precision accuracy after just a small number of nodes. Both methods exhibit nearly identical convergence behavior, with Gauss–Legendre achieving the plateau slightly earlier.

In contrast, the trapezoidal rule converges much slower and again remains much less accurate even for large n. This confirms the clear advantage of spectral-type methods for smooth polynomial functions.

Task 5 – Discussion

The Trapezoidal Rule is simple, but it converges slowly and is sensitive to oscillations, which limits its accuracy for smooth functions, but as we saw - at the same time gives this method an advantage for oscillatory functions.

Gauss–Legendre quadrature provides very high accuracy and fast convergence for smooth integrands, but its performance can degrade for non-smooth functions plus this method requires computing non-uniform nodes and weights.

Clenshaw–Curtis quadrature offers accuracy comparable to Gauss–Legendre while being easier to implement and more numerically stable due to its use of Chebyshev nodes and FFT-based techniques.

Overall, trapezoidal methods are reliable but inefficient, while Gauss–Legendre and Clenshaw–Curtis are highly efficient for smooth function although lose the reliabilty when regularity is lost.